

data cannot be directly used for training; it needs to be cleaned. If data is not good, a considerable amount of time needs to be spent to pre-process it so that it can be used. Data preprocessing takes a lot of time and effort to generate a quality data set. Hence data sets play a predominant role in training of the machine learning and deep learning models.

A quality data set saves the time and effort needed for machine learning and deep learning researchers and data scientists. A few years back everyone would generate data from different sources required for the learning algorithms. There were no standard data sets available and these were stored in private and public locations of the user's choice. It was difficult to generalise the insights, make developments to learning algorithms, and enable data sets to be explored by wider audiences. As machine learning and deep learning technologies advanced, their frameworks started providing built-in data sets as part of their packages. This would enable users to access similar data sets, save the effort and time required for data preprocessing, and direct their knowledge to building high performing and complex models that provided advanced insights. Today, machine learning and deep learning frameworks like Scikit-learn, TensorFlow, and PyTorch (among many others) provide built-in data sets for almost all types of data like text, image, audio and video, with sufficient samples for training, validation and testing of models.

Libraries with built-in data sets

Lots of libraries nowadays come with built-in data sets that can be used to train machine learning algorithms. Built-in data sets prove to be very useful when it comes to practising ML algorithms — you need some random, yet sensible data to apply the techniques and get your hands dirty. Many modules in Python contain some common data sets similar to the popular 'Iris' data, MNIST digits data and Boston housing price data set.

Scikit-learn comes with a few small standard data sets that do not require one to download any file from some external website. Similarly, TensorFlow provides the TensorFlow data set for use. PyTorch also provides a lot of data sets for use in machine learning and deep learning. There are many data sets available but in this article, we will only discuss the usage of TensorFlow and PyTorch data sets.

Types of data sets available

There are different types of data sets available, as listed below. The type of data set used depends on the problem at hand.

Audio data: This is basically useful in machine learning and deep learning tasks such as speech recognition and emotion recognition. Some audio data is also used for identification of diseases like Alzheimer's and Covid-19 (the audio data of people coughing can be used to identify Covid-19). The speech data of people talking is also used to classify those with Alzheimer's disease.

Image data: This is used for classification and object detection. Image data available with TensorFlow such as COCO and Wider-face can be used for object detection. Wider-face is a data set of images with pictures of people, and it can be used in deep learning to identify faces. Image data can also be used for deep learning algorithms such as segmentation, which can be used for self-driving cars.

Text data: Text data can be used for various natural language processing tasks such as movie review classification, fake news detection, summarisation, answering questions, topic detection, transcript summarisation, and detection of action items from emails, etc. An example of text data is the collection of email messages sent by employees of Enron Corporation.

Video data: Video data can be used for object detection in videos and for video segmentation. An example is the DAVIS (Densely Annotated Video Segmentation) data set from TFDS (TensorFlow data set).

Translation data sets: Labelled data is also available for translation. These data sets can be used for machine translation tasks for different languages.

Given below are a few of the standard built-in data sets that are provided as part of the deep learning and machine learning frameworks. Each framework provides a superset of these data sets. The data sets listed below showcase the diversity provided by these frameworks so that users can exploit them to their advantage for building models catering to their domain.

- **CelebA** (CelebFaces Attributes) is a large-scale face attributes data set with more than 200,000 celebrity images, each with 40 attribute annotations. It can be used for face attribute recognition, face detection and landmark (or facial part) localisation.
- **CIFAR-10** (Canadian Institute for Advanced Research) data set consists of 60,000 32x32 colour images, categorised into 10 classes with 6000 images per class. It can be used for image classification and computer vision tasks.
- **Cityscapes** data set consists of 25,000 samples of video sequences recorded in street scenes, with pixel-level annotations. It can be used for classification and object detection.
- **COCO** (Common Objects in Context) data set consists of images of daily scenes of common objects in their natural context. It has around 2.5 million images with labelled text which can be used for object recognition, classification and caption generation.
- **Fashion-MNIST** (Modified National Institute of Standards and Technology) data set consists of images similar to the MNIST data set but from fashion product databases. It has around 60,000 images and can be used for image classification.
- **HMDB** (Human Metabolome Database) is collected from a variety of sources, most of which